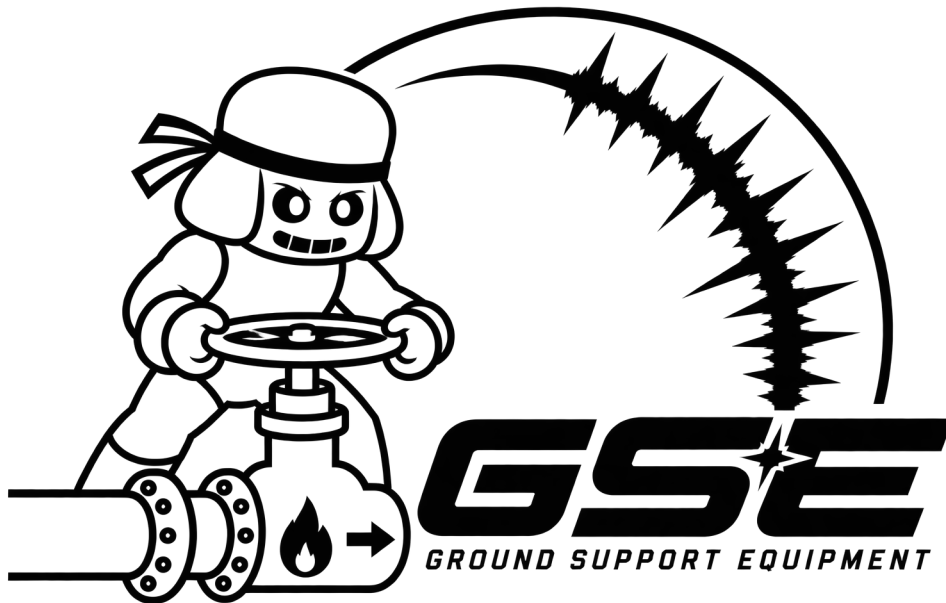


VALVE POWER BOARD

RUBY - REV A



RESPONSIBLE ENGINEER:
NICHOLAS ALVES

REVIEWERS:

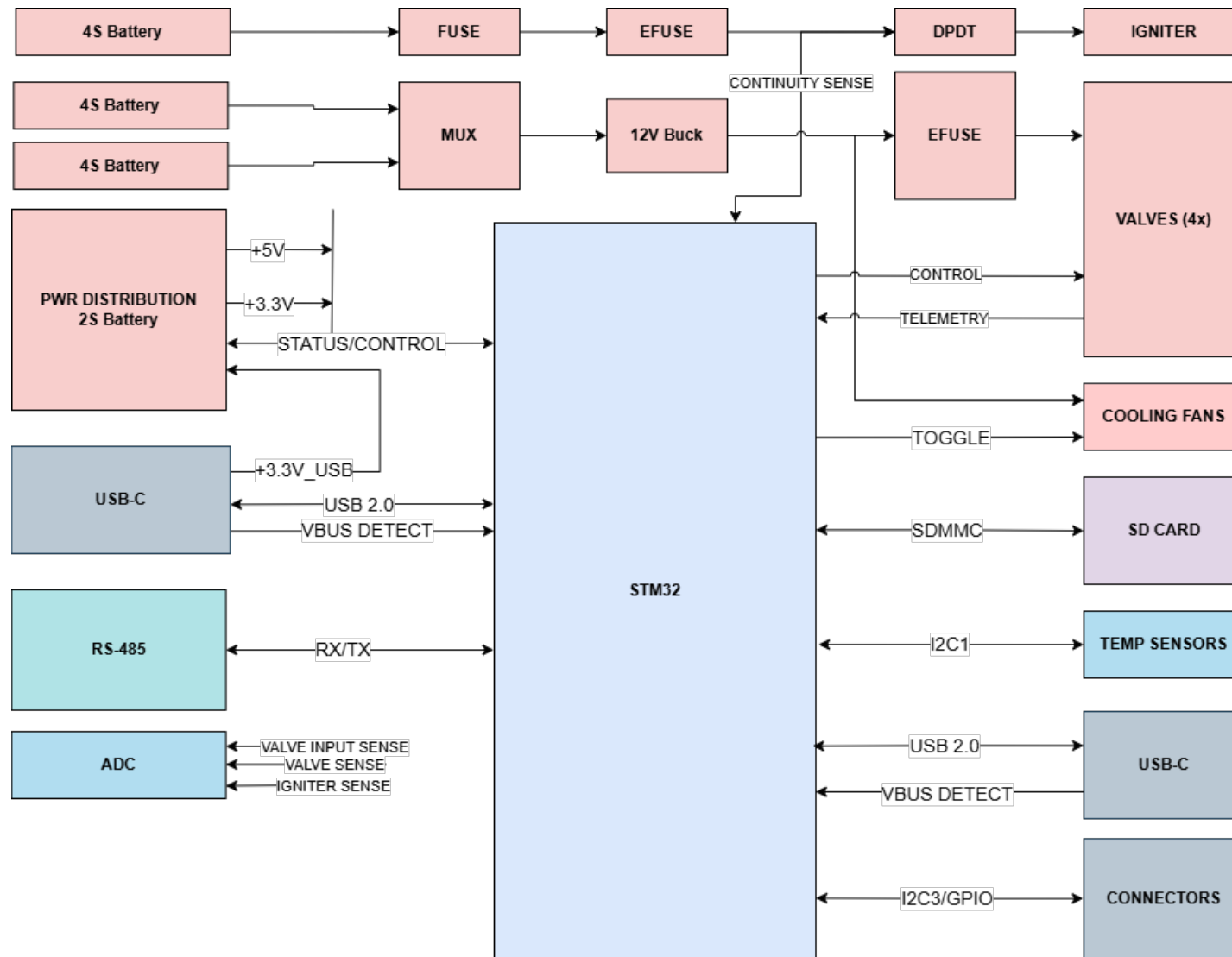
SCRT
STUDENT COMPETITION ROCKETRY TEAM

Sheet: /Title Page/
File: top-level.kicad_sch

Title: TITLE PAGE

Size: A4 Date: 2026-10-08
KiCad E.D.A. 10.0.5

Rev: A
Id: 1/32



Sheet: /Block Diagrams/
File: block_diagrams.kicad_sch

Title: BLOCK DIAGRAM

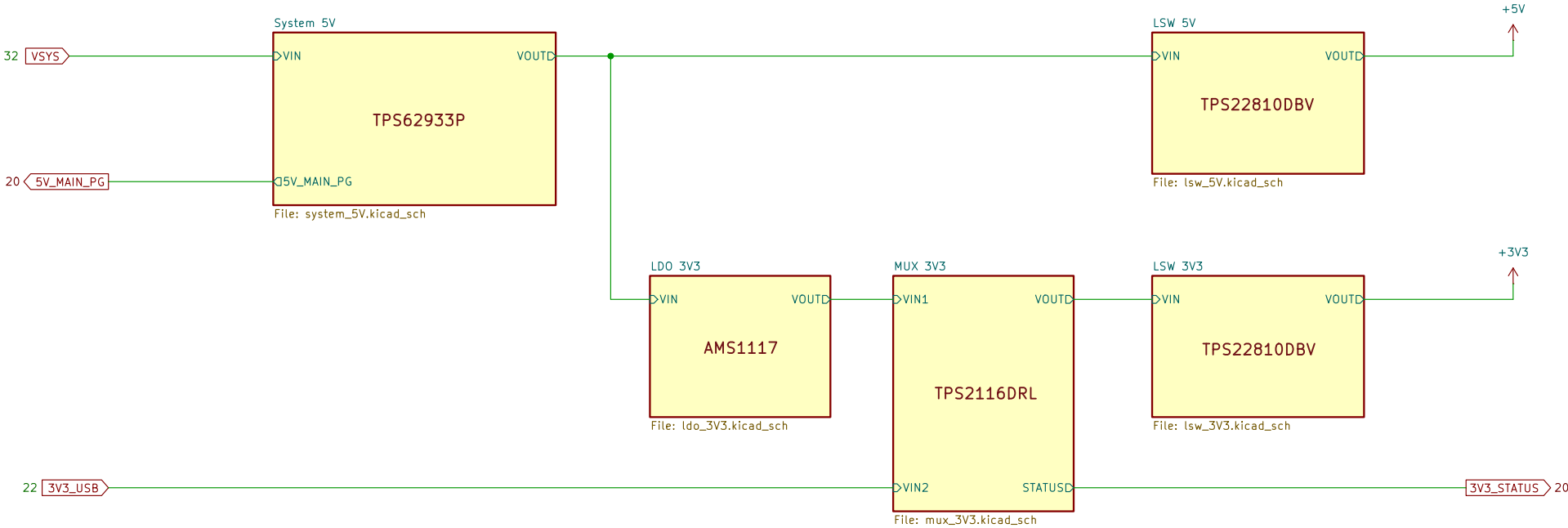
Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A

Id: 2/32

SYSTEM POWER



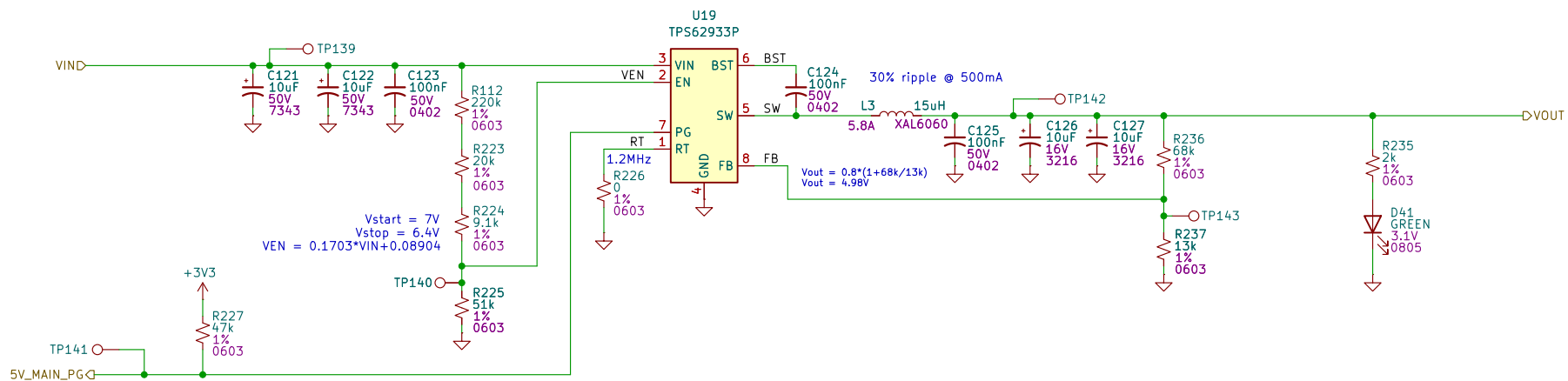


STUDENT COMPETITION ROCKETRY TEAM

Sheet: /System Power/
File: system_power.kicad_sch

Title: SYSTEM POWER

Size: A4	Date: 2026-10-08	Rev: A
KiCad E.D.A. 10.0.5		Id: 3/32



Sheet: /System Power/System 5V/
File: system_5V.kicad_sch

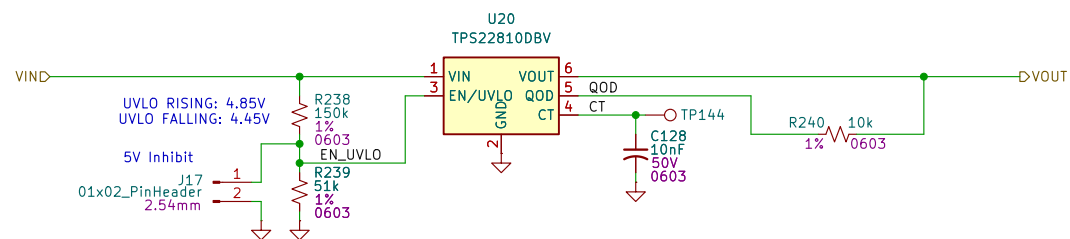
Title: SYSTEM 5V

Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A

Id: 4/32



Sheet: /System Power/LSW 5V/
File: lsw_5V.kicad_sch

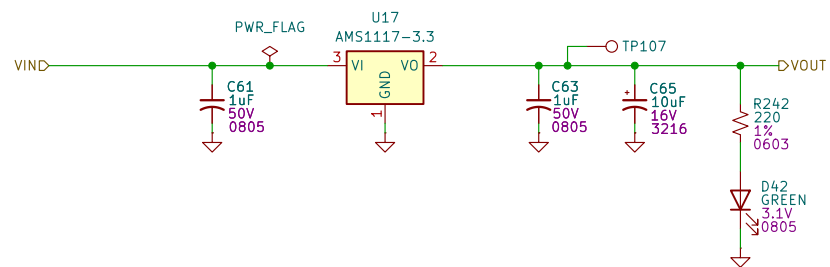
Title: LSW 5V

Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A

Id: 5/32



Sheet: /System Power/LDO 3V3/
File: ldo_3V3.kicad_sch

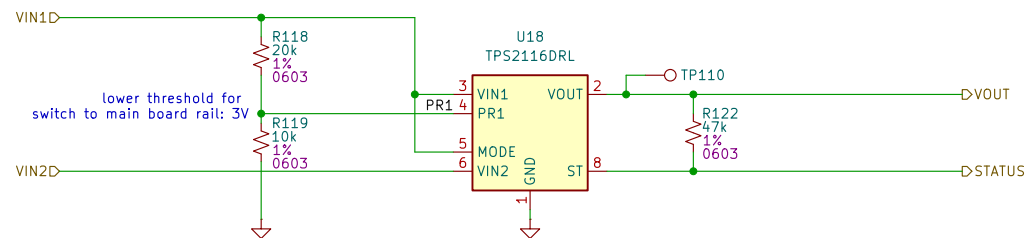
Title: LDO 3V3

Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A

Id: 6/32



Sheet: /System Power/MUX 3V3/
File: mux_3V3.kicad_sch

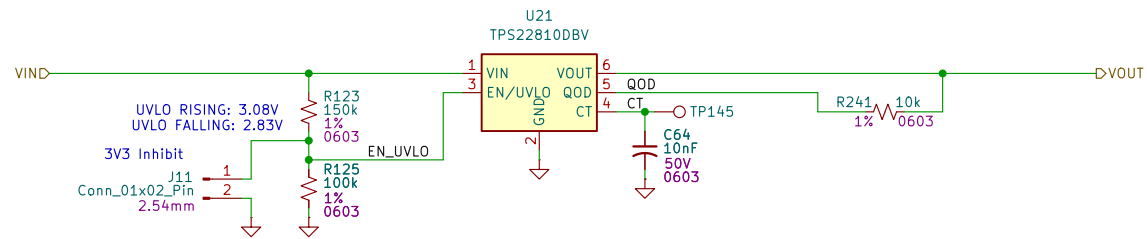
Title: MUX 3V3

Size: A4 Date: 2026-10-08

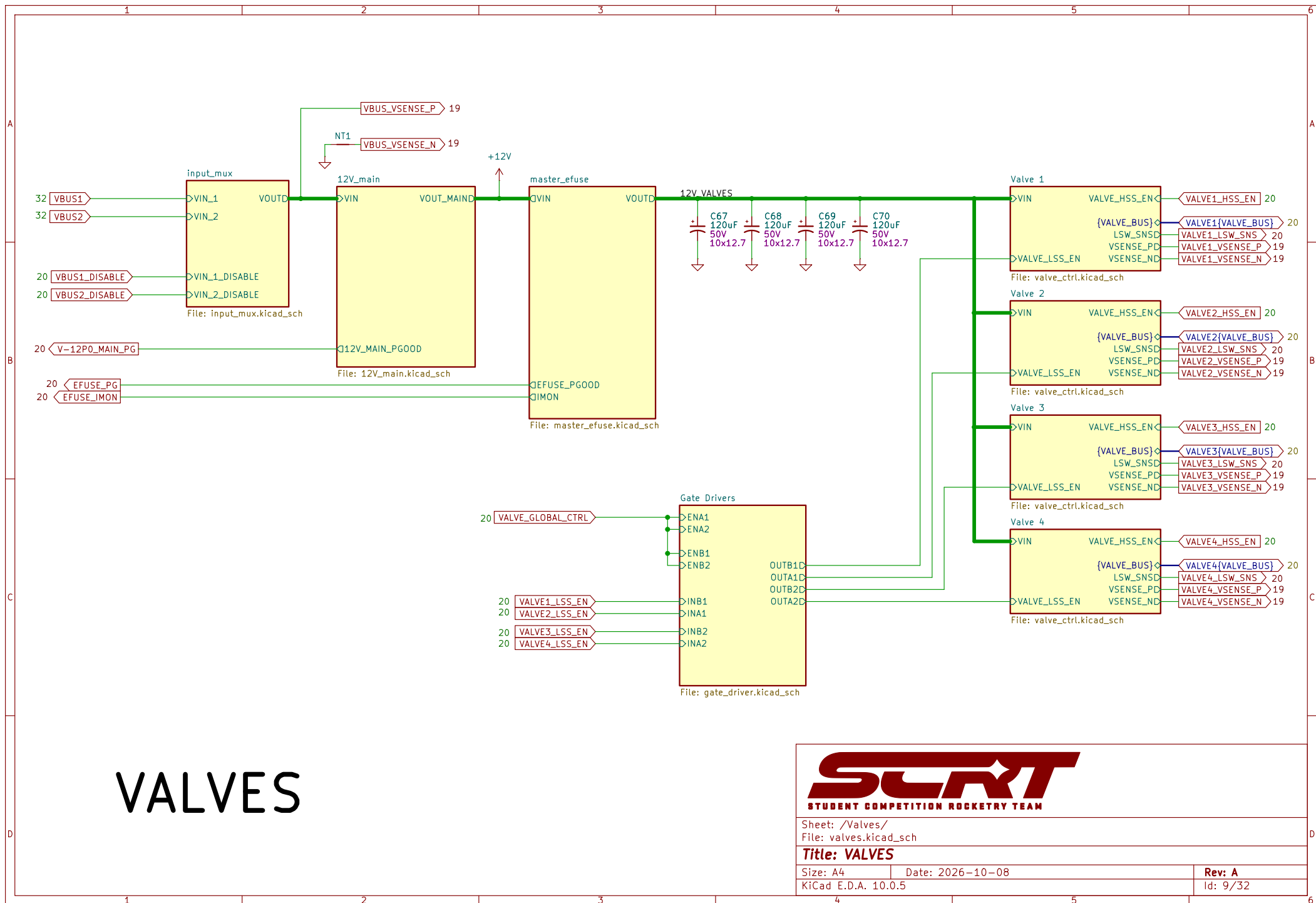
KiCad E.D.A. 10.0.5

Rev: A

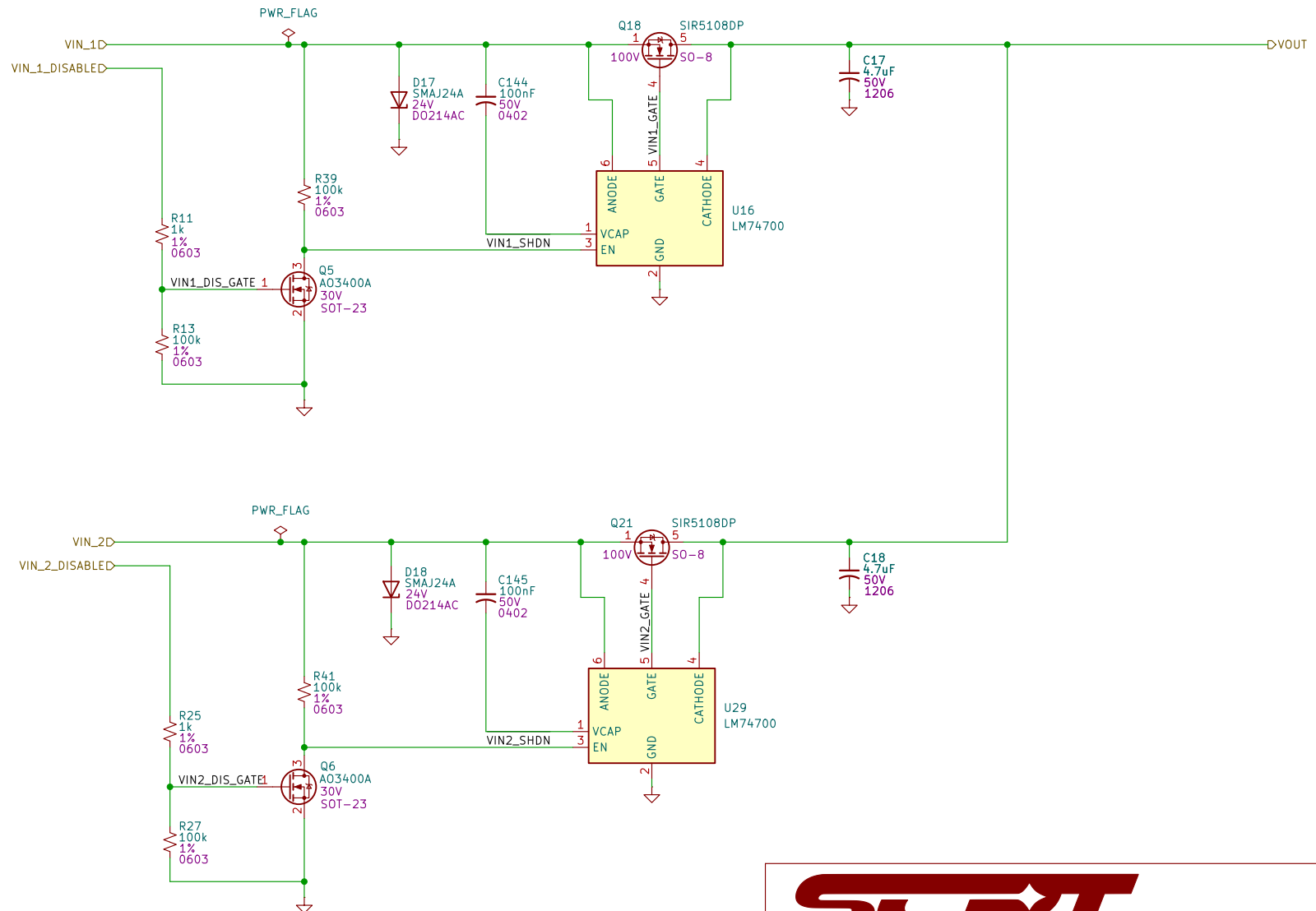
Id: 7/32



Sheet: /System Power/LSW 3V3/ File: lsw_3V3.kicad_sch		
Title: LSW 3V3		
Size: A4	Date: 2026-10-08	Rev: A
KiCad E.D.A. 10.0.5		Id: 8/32



Protect against MOSFET avalanche



Sheet: /Valves/input_mux/
File: input_mux.kicad_sch

Title: INPUT MUX

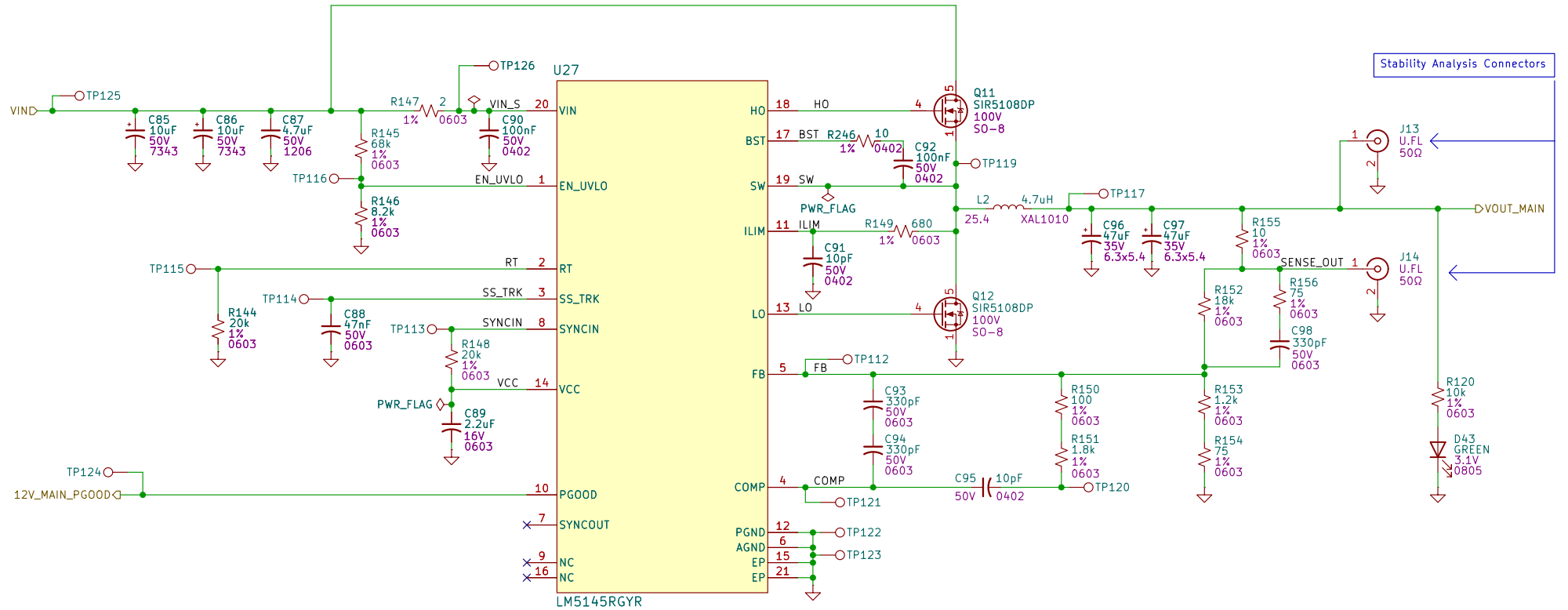
Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A

Id: 10/32

12V MAIN



Sheet: /Valves/12V_main/
File: 12V_main.kicad_sch

Title: 12V MAIN

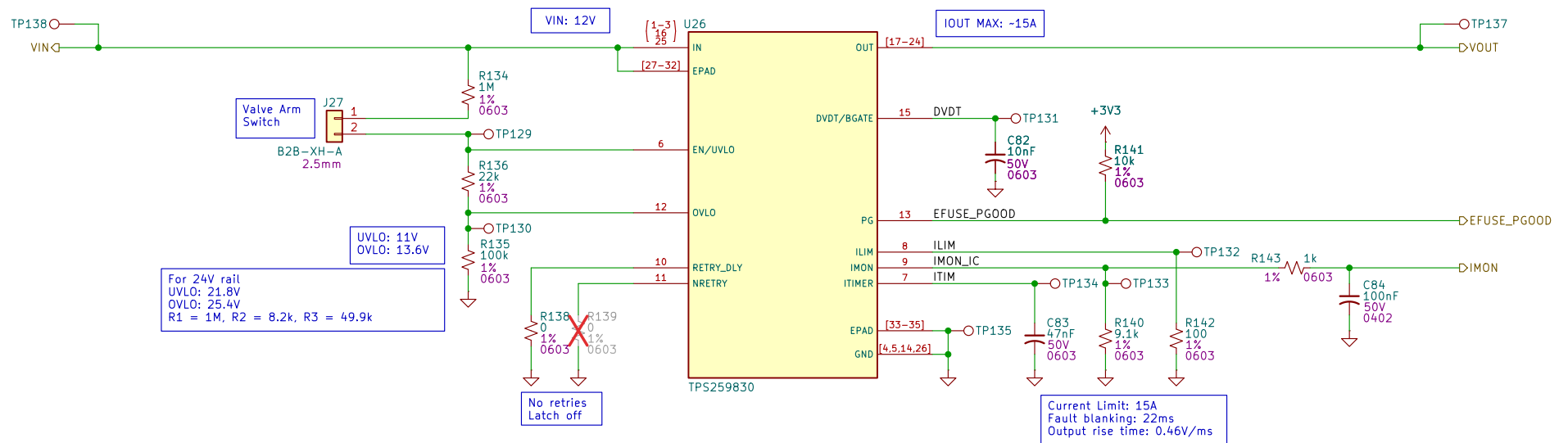
Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A

Id: 11/32

EFUSE

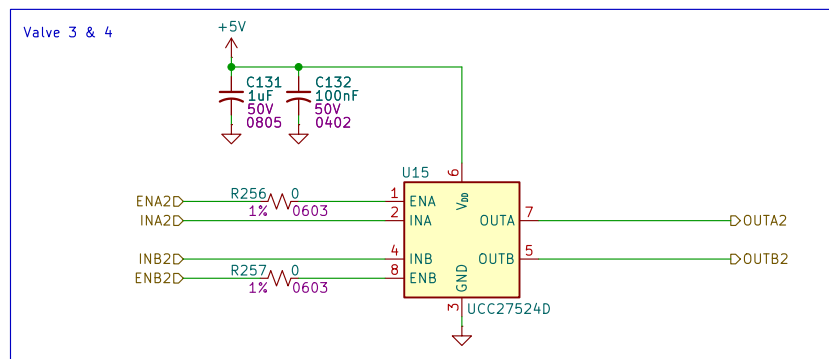
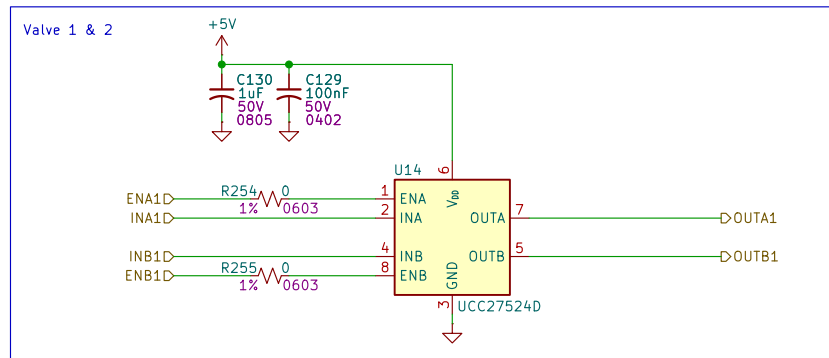


Sheet: /Valves/master_efuse/
File: master_efuse.kicad_sch

Title: EFUSE

Size: A4 Date: 2026-10-08
KiCad E.D.A. 10.0.5

Rev: A
Id: 12/32



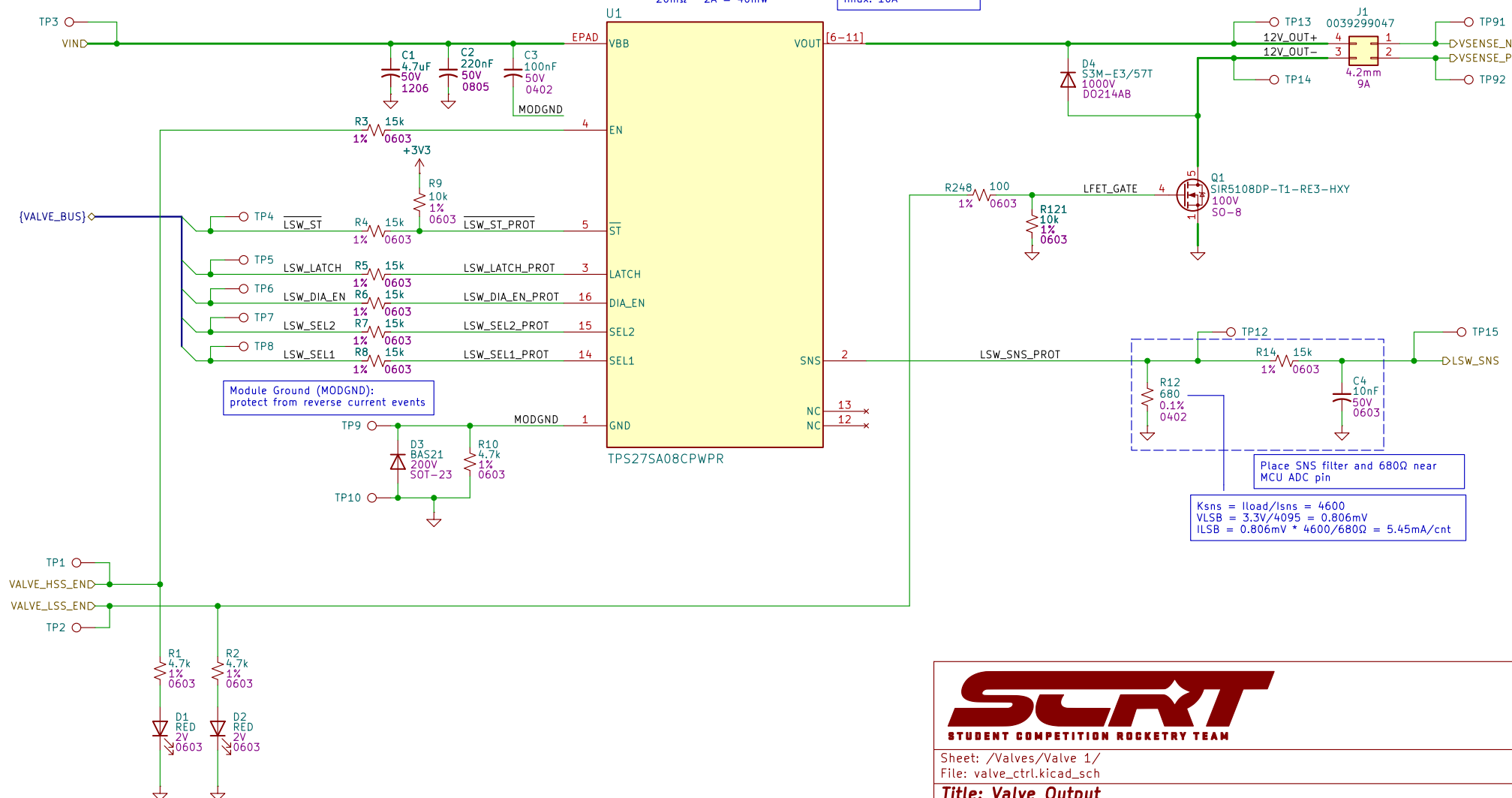
VALVE OUTPUT POWER

Valve Enable Procedure:

1. Enable common ELOAD (U26)
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)

20mΩ * 2A = 40mW

12V, 2A nominal output
Vmax: 36V
Imax: 10A



SCRT
STUDENT COMPETITION ROCKETRY TEAM

Sheet: /Valves/Valve 1/
File: valve_ctrl.kicad_sch

Title: Valve Output

Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A

Id: 14/32

Valve Enable Procedure:

1. Enable common ELOAD (U26)
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)



File: valve_ctrl.kicad_sch

Title: Valve Output

Size: A4	Date:
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KiCad E.D.A. 10.0.5

Date: 2026-10-08

Rev: A

Id: 15/32

Valve Enable Procedure:

1. Enable common ELOAD (U26)
2. Enable low-side switch (LSS)
3. Enable high-side switch (HSS)

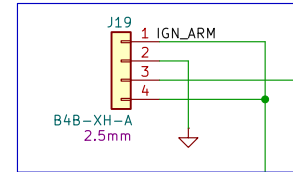


Title: Valve Output

Id: 16/32

IGNITER SYSTEM

ARMING SWITCH
EXTERNAL LED ARMED
CONNECTOR



D33 clamps Q8
Vsg at -15V
IOUT MAX: 14.7A

Current Limit: 14.7A
Fault blanking: 22ms
Output rise time: 0.46V/ms

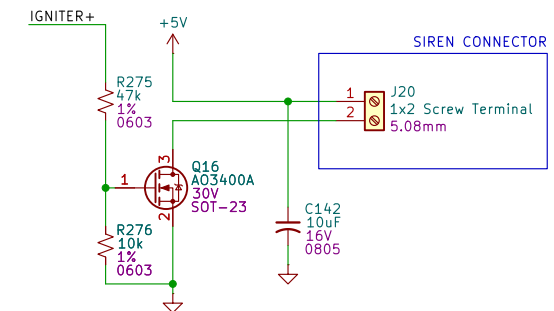
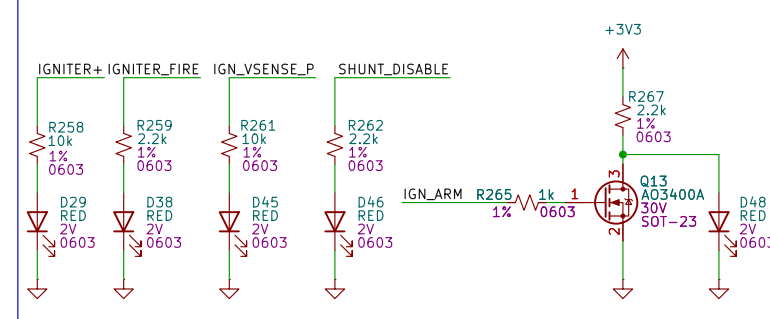
Continuity check:
1. ELOAD disabled, Q8 disarmed.
2. K1 shorts pins 1-3.
3. MCU checks continuity_sense net:
i. If HIGH, igniter not present
ii. If LOW, igniter present

sense @ 12V: 1.69V
sense @ 16V: 2.25V
sense @ 20V: 2.82V

TEST POINTS

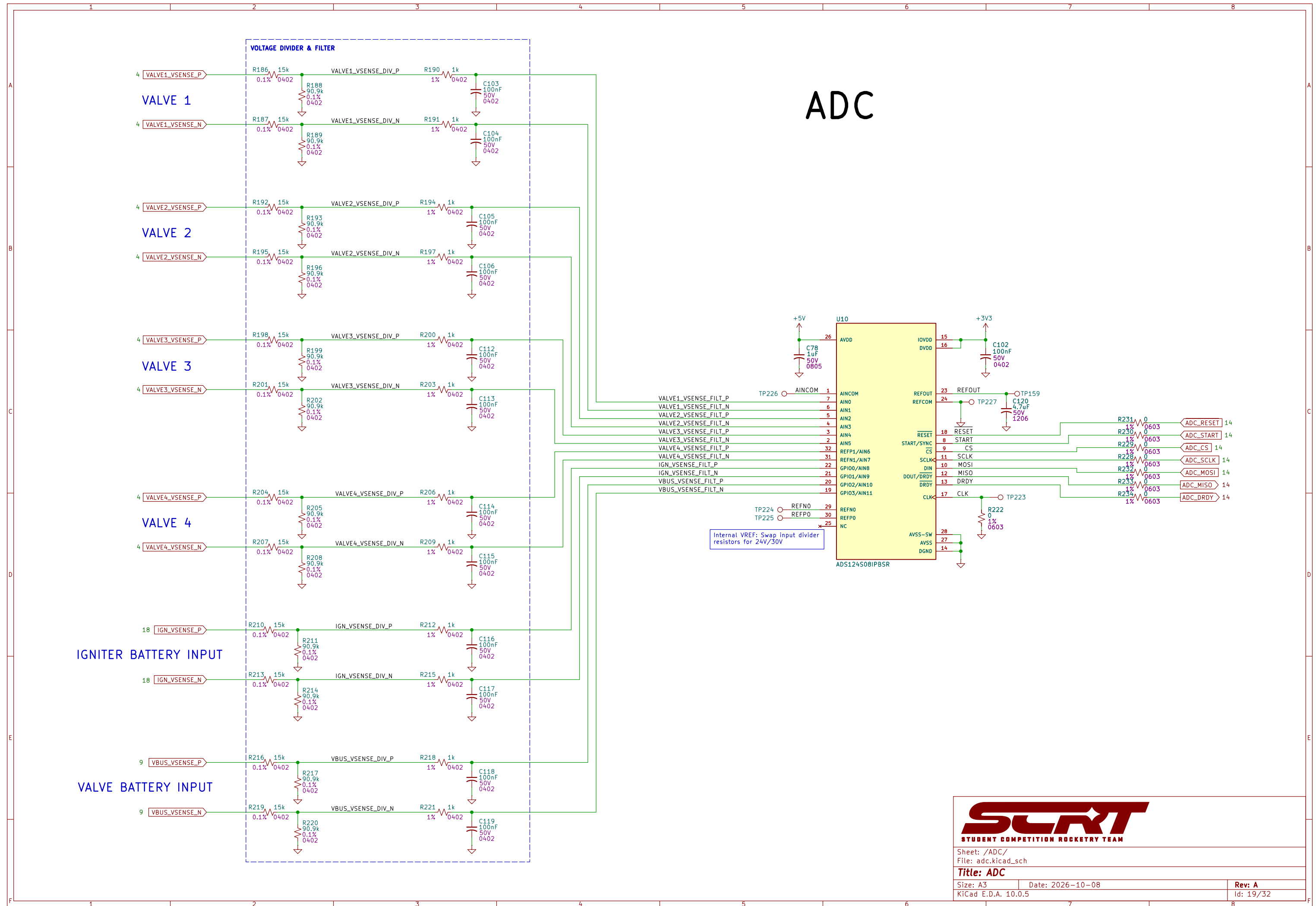
TP41 TestPoint	IGN_VSENSE_P	TP67 TestPoint	IMON
TP56 TestPoint	IGNITER_FIRE	TP68 TestPoint	ITIM
TP61 TestPoint	EFUSE_OVLO	TP69 TestPoint	SENSE
TP62 TestPoint	SHUNT_DISABLE	TP70 TestPoint	+5V
TP63 TestPoint	IGNITER+	TP71 TestPoint	IGN_RLY-
TP64 TestPoint	DVDT		
TP65 TestPoint	PGOOD_IGNITER		
TP66 TestPoint	ILIM		

LEDs

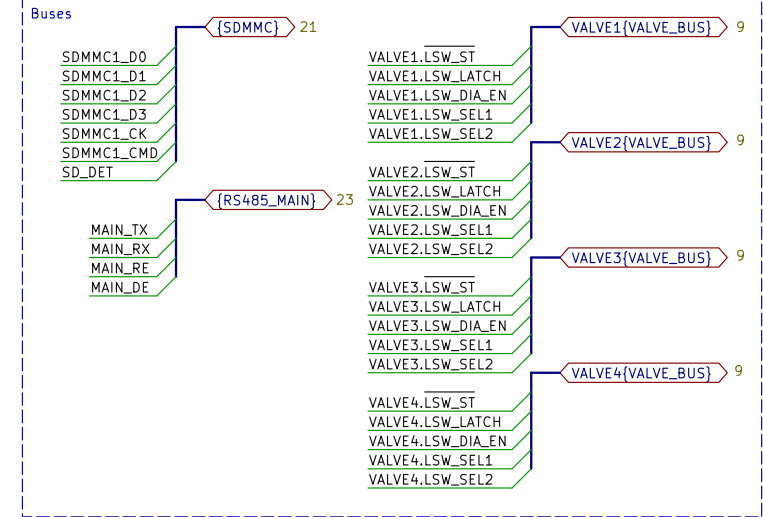
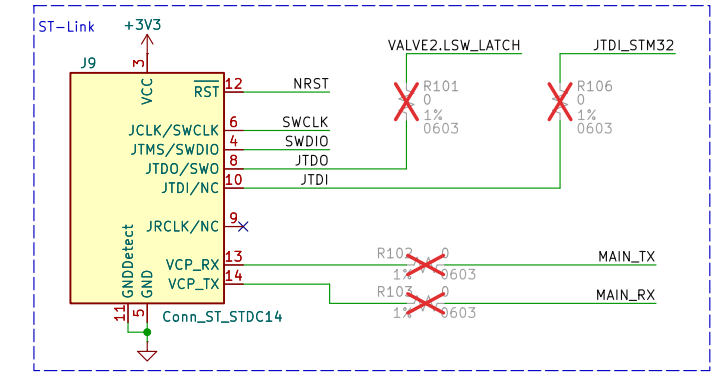
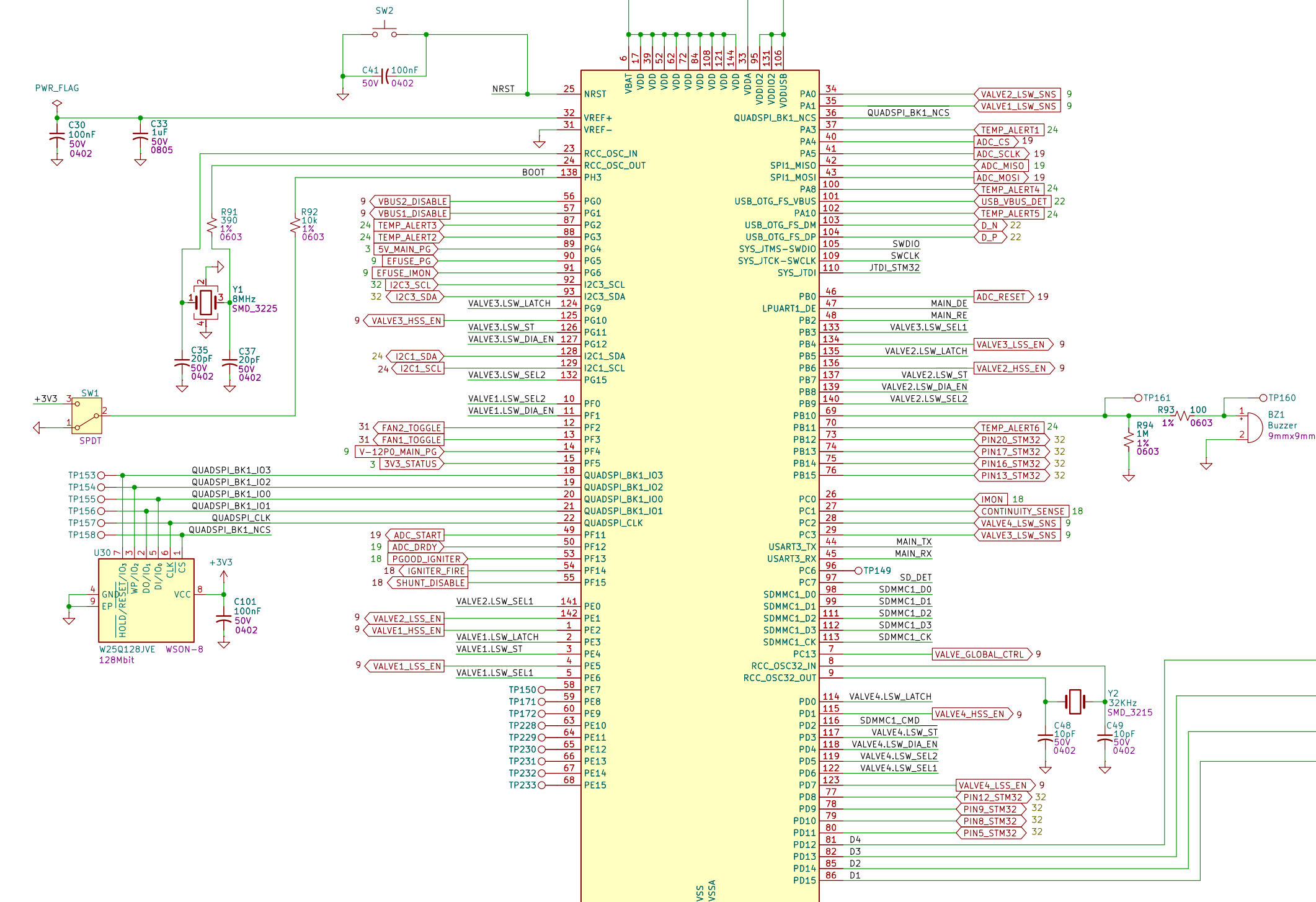
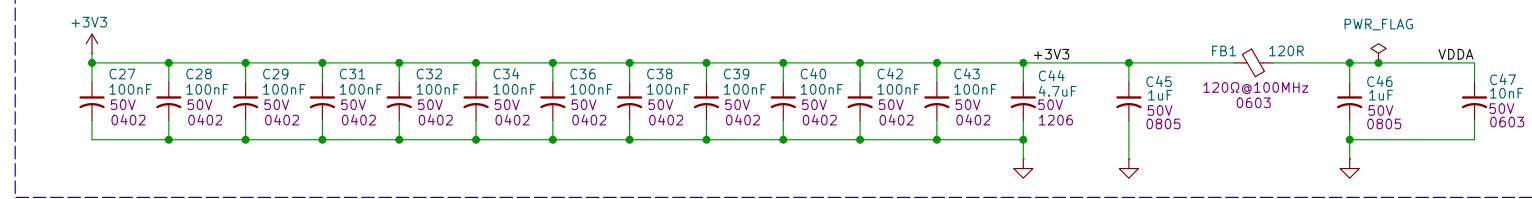


Sheet: /Igniter/ File: igniter.kicad_sch		
Title: IGNITER SYSTEM		
Size: A3	Date: 2026-10-08	Rev: A
KiCad E.D.A. 10.0.5		Id: 18/32

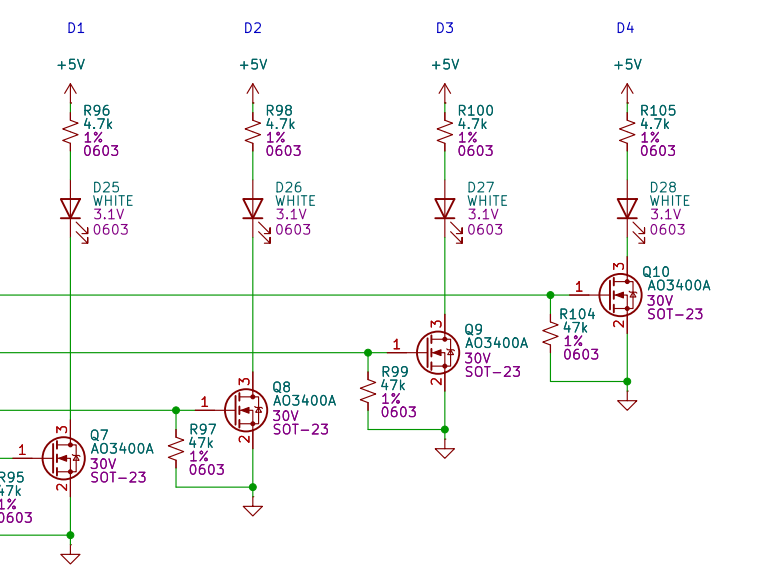
ADC



Supply & Analog Supply Filtering



Debugging and Status LEDs



STM32L496ZGT6



STUDENT COMPETITION ROCKETRY TEAM

Sheet: /STM32/
File: stm32.kicad_sch

Title: STM32L496ZGT6

Size: A3 Date: 2026-10-08 **Rev: A**

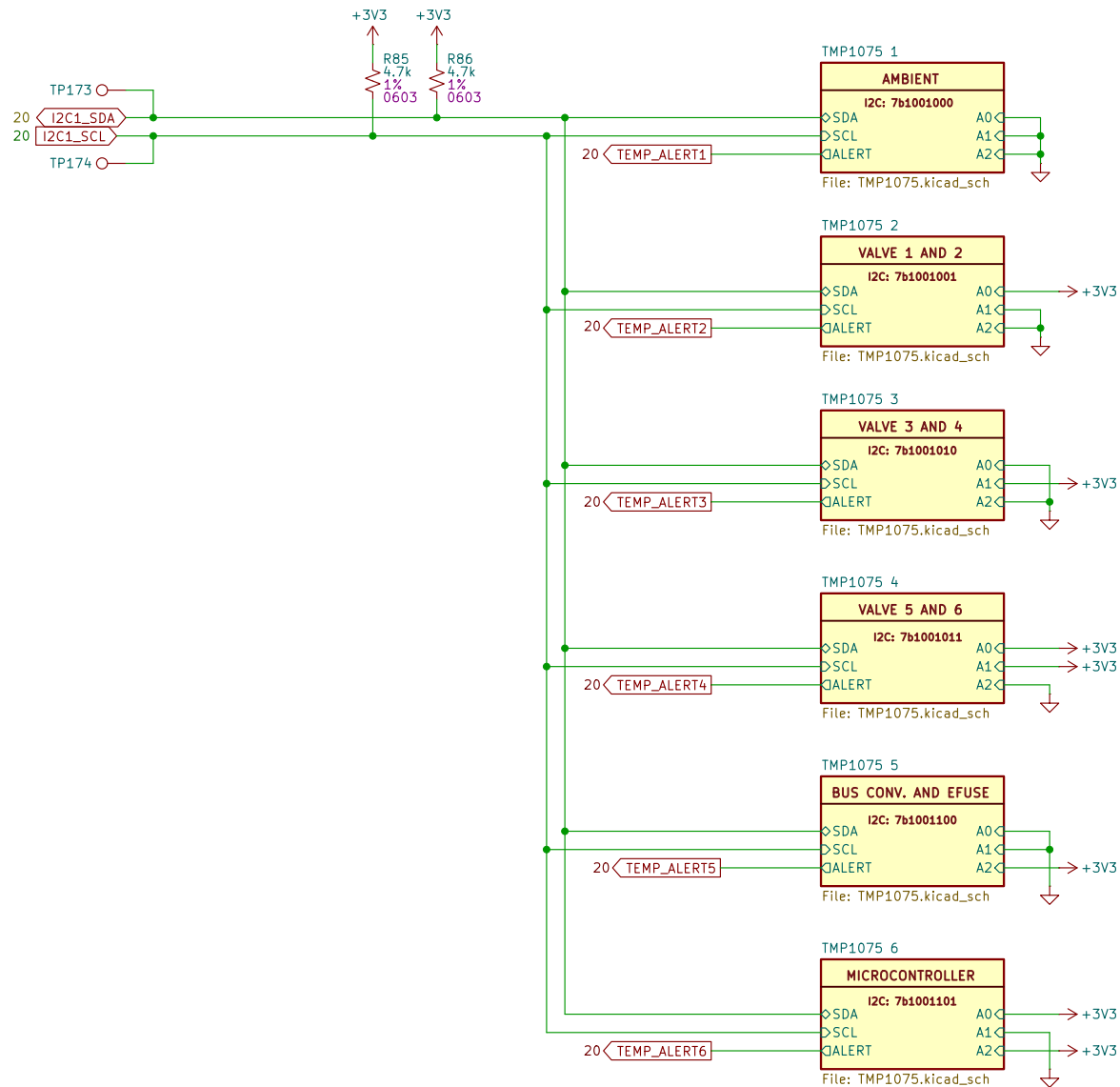
KiCad E.D.A. 10.0.5 Id: 20/32

[illegible]

[illegible]

[illegible]

TEMPERATURE SENSORS



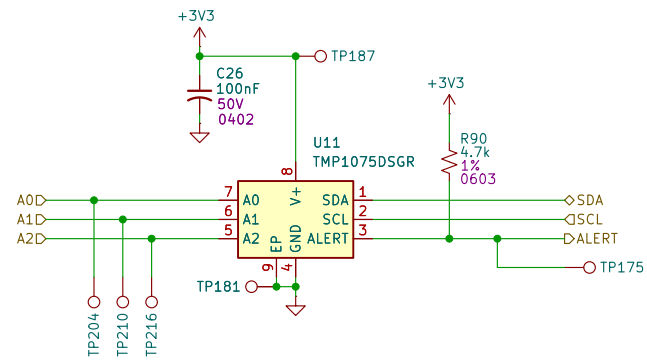
Sheet: /Sensors/
File: sensors.kicad_sch

Title: SENSORS

Size: A4 Date: 2026-10-08
KiCad E.D.A. 10.0.5

Rev: A
Id: 24/32

TMP1075



Sheet: /Sensors/TMP1075 1/
File: TMP1075.kicad_sch

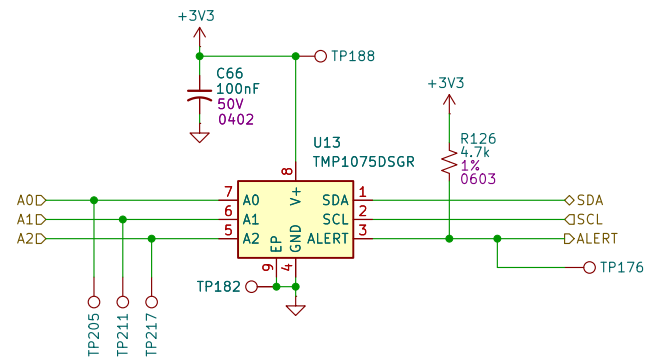
Title: TMP1075

Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A
Id: 25/32

TMP1075



Sheet: /Sensors/TMP1075 2/
File: TMP1075.kicad_sch

Title: TMP1075

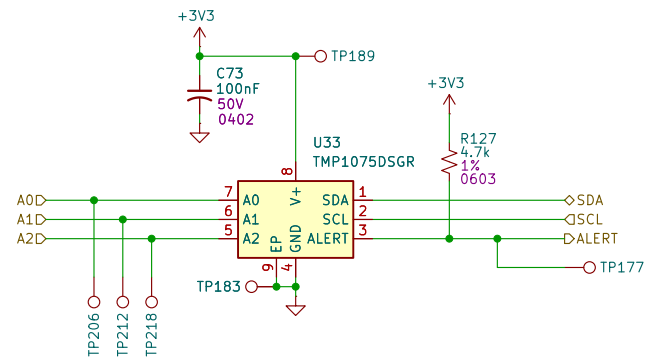
Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A

Id: 26/32

TMP1075



Sheet: /Sensors/TMP1075 3/
File: TMP1075.kicad_sch

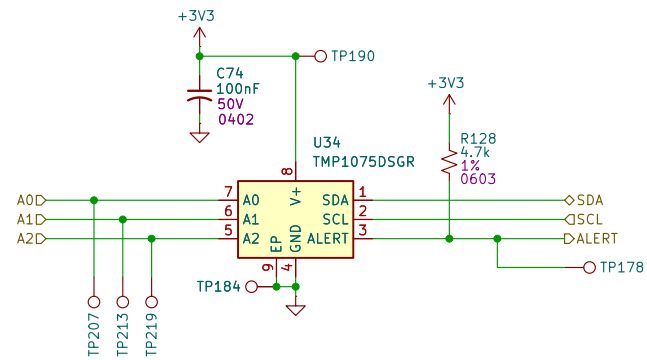
Title: TMP1075

Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A
Id: 27/32

TMP1075



Sheet: /Sensors/TMP1075 4/
File: TMP1075.kicad_sch

Title: TMP1075

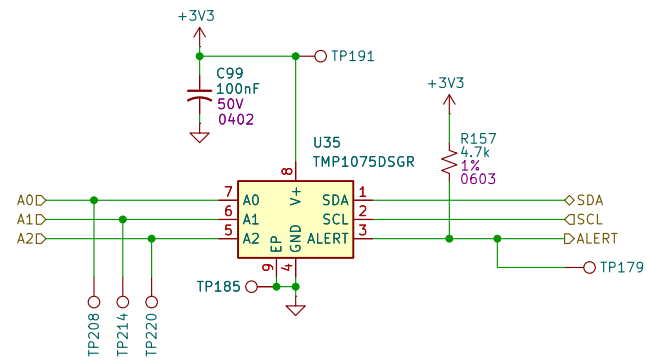
Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A

Id: 28/32

TMP1075



Sheet: /Sensors/TMP1075 5/
File: TMP1075.kicad_sch

Title: TMP1075

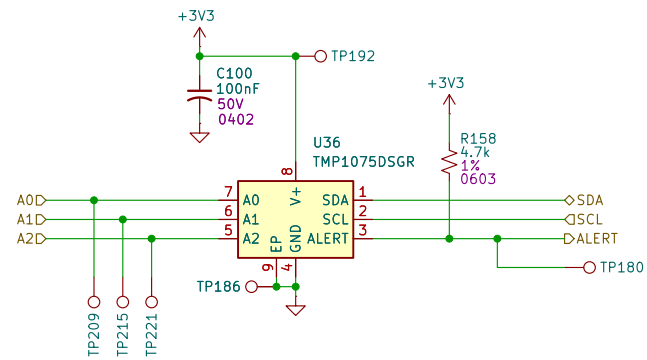
Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A

Id: 29/32

TMP1075



Sheet: /Sensors/TMP1075 6/
File: TMP1075.kicad_sch

Title: TMP1075

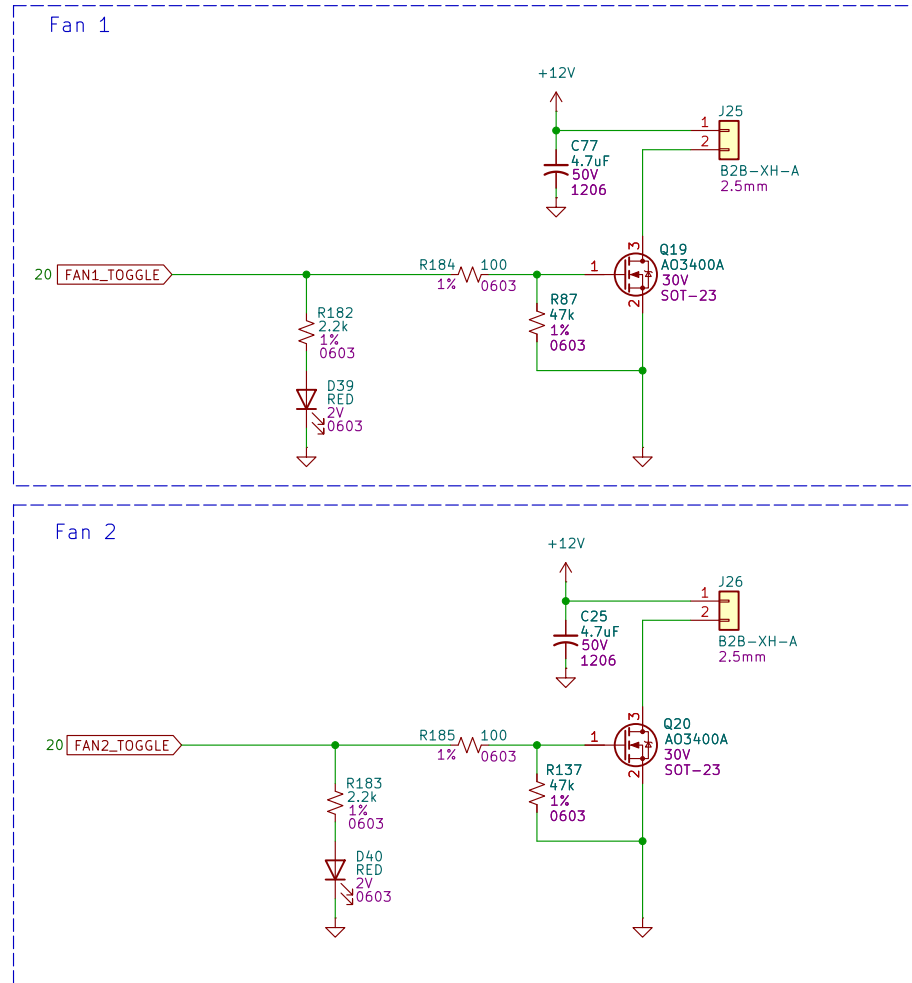
Size: A4 Date: 2026-10-08

KiCad E.D.A. 10.0.5

Rev: A

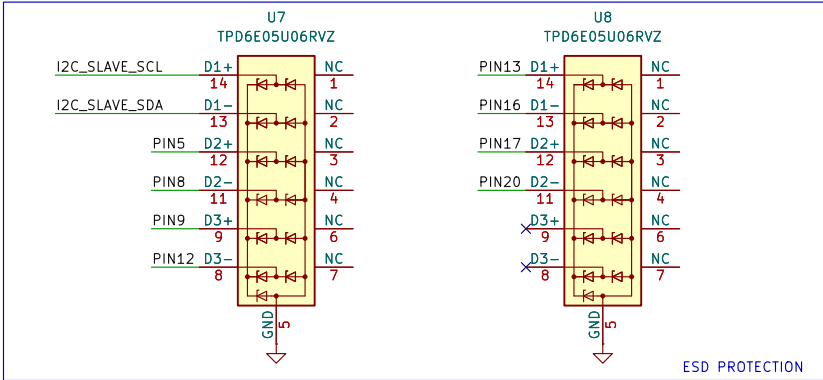
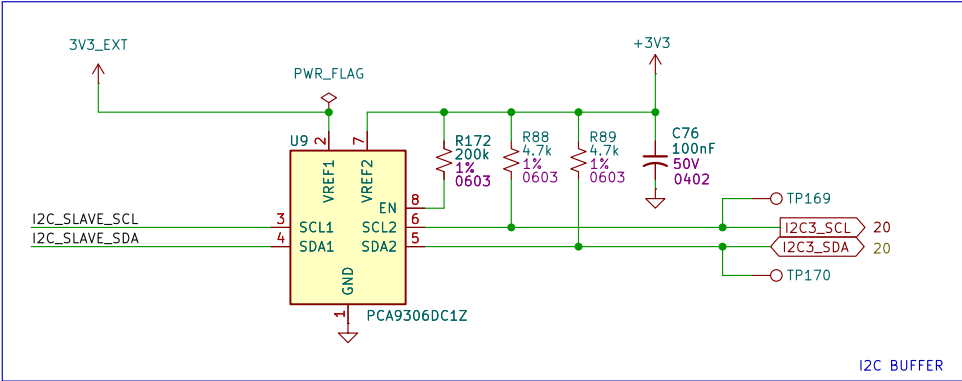
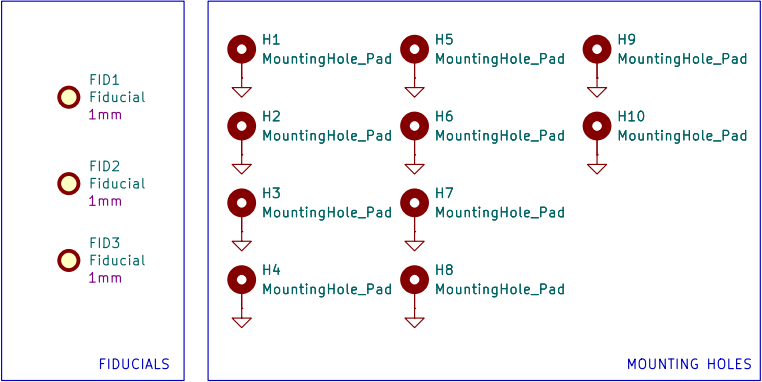
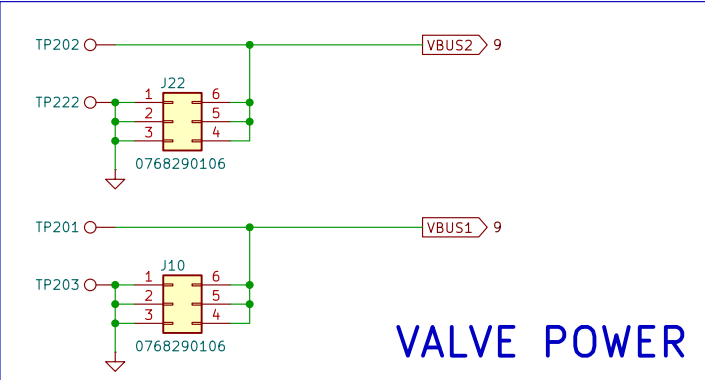
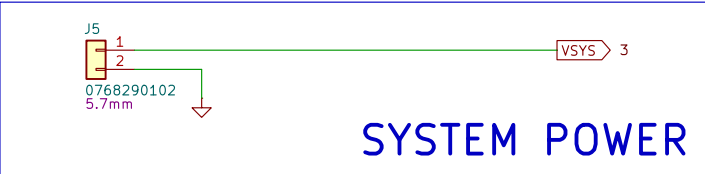
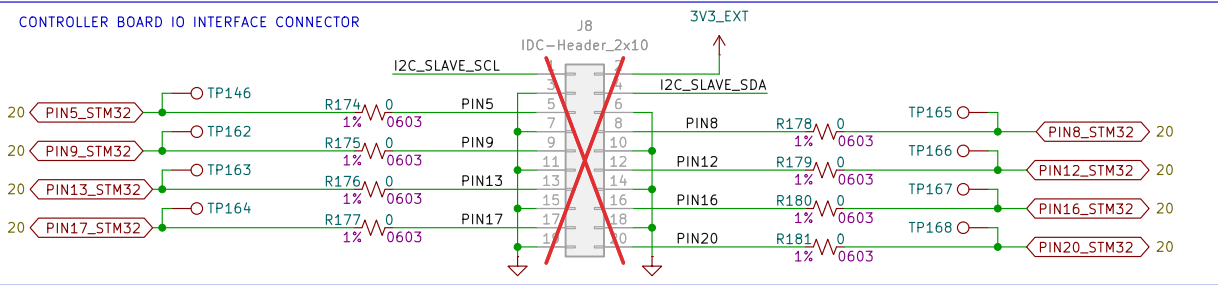
Id: 30/32

FANS



Sheet: /Fans/ File: fans.kicad_sch		
Title: Fans		
Size: A4	Date: 2026-10-08	Rev: A
KiCad E.D.A. 10.0.5	Id: 31/32	

MECHANICAL



Sheet: /Connectors/
File: connectors.kicad_sch

Title: MECHANICAL

Size: A4
KiCad E.D.A. 10.0.5

Date: 2026-10-08

Rev: A
Id: 32/32